SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

REVISION EXAMINATION - 2025

MATHEMATICS

Time Allowed : 3.00 Hours] **[Max. Marks : 90**

PART - I

20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. Let X = {1, 2, 3, 4}, Y = {a, b, c, d} and f = {(1, a), (4, b), (2, c), (3, d), (2, d)}. Then f is

(1) an one-to-one function (2) an onto function

(3) a function which is not one-to-one (4) not a function

2. If 3 is the logarithm of 343 , then the base is

(1) 5 (2) 7 (3) 6 (4) 9

3. If $f(\theta) = |\sin\theta| + |\cos\theta|, \theta \in R$, then $f(\theta)$ is in the interval

(1) [0,2] (2) $[1,\sqrt{2}]$ (3) [1,2] (4) [0,1]

4. In a plane there are 10 points are there out of which 4 points are collinear, then the number of triangles formed is

(1) 110 (2) $^{10}C_3$ (3) 120 (4) 116

5. If $^nC_4$, $^nC_5$, $^nC_6$ are in AP the value of n can be

(1) 14 (2) 11 (3)9 (4) 5

6. If a is the arithmetic mean and g is the geometric mean of two numbers, then

(1) $a \le g$ (2) $a \ge g$ (3) $a = g$ (4) $a > g$.

7. If the point (8, −5) lies on the locus $\frac{x^2}{16} - \frac{y^2}{25} = k$, then the value of k is

(1) 0 (2) 1 (3) 2 (4) 3

8. If a vertex of a square is at the origin and its one side lies along the line $4x + 3y - 20 = 0$, then the area of the square is

(1) 20 sq. units (2) 16 sq. units (3) 25 sq. units (4) 4 sq.units

9. If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in geometric progression with the same common ratio, then the points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are

(1) vertices of an equilateral triangle (2) vertices of a right angled triangle

(3) vertices of a right angled isosceles triangle (4) collinear

10. Which one of the following is not true about the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}$?

(1) a scalar matrix (2) a diagonal matrix

(3) an upper triangular matrix (4) a lower triangular matrix

11. The value of $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{DA} + \overrightarrow{CD}$ is

(1) $\overrightarrow{AD}$ (2) $\overrightarrow{CA}$ (3) $\vec{0}$ (4) $-\overrightarrow{AD}$

12. If $\vec{r} = \frac{9\vec{a}+7\vec{b}}{16}$, then the point P whose position vector $\vec{r}$ divides the line joining the points with position vectors $\vec{a}$ and $\vec{b}$ in the ratio

(1) 7:9 internally (2) 9:7 internally (3) 9:7 externally (4) 7:9 externally

13. If $\lim_{x\to 0}\frac{\sin px}{\tan 3x} = 4$, then the value of p is

(1) 6 (2) 9 (3) 12 (4) 4

14. If $f(x) = \begin{cases} x-5 & \text{if } x \le 1 \\ 4x^2-9 & \text{if } 1 < x < 2 \\ 3x+4 & \text{if } x \ge 2 \end{cases}$, then the right hand derivative of f(x) at x = 2 is

(1) 0 (2) 2 (3) 3 (4) 4

15. $\int \frac{e^{6\log x}-e^{5\log x}}{e^{4\log x}-e^{3\log x}}dx$ is

(1) $x+c$ (2) $\frac{x^3}{3}+c$ (3) $\frac{3}{x^3}+c$ (4) $\frac{1}{x^2}+c$

16. If A and B are two events such that $A \subset B$ and $P(B) \neq 0$, then which of the following is correct?

(1) $P(A/B) = \frac{P(A)}{P(B)}$ (2) $P(A/B) < P(A)$ (3) $P(A/B) \geq P(A)$ (4) $P(A/B) > P(B)$

17. The logarithmic expression $\log_a b \cdot \log_b c \cdot \log_c a$ simplifies to:

(1) 1 (2) 0 (3) abc (4) a^bc

18. $\lim_{x\to 0}\frac{\sin^3 x}{x^3}$ is:

(1) 1 (2) $\frac{1}{3}$ (3) 3 (4) 0

19. The value of $|\vec{a}\times\vec{b}|^2 + (\vec{a}\cdot\vec{b})^2$ is:

(1) $|\vec{a}|^2+|\vec{b}|^2$ (2) $|\vec{a}|^2|\vec{b}|^2$ (3) $|\vec{a}|^2|\vec{b}|^2 + (\vec{a}\cdot\vec{b})$ (4) $|\vec{a}|^2|\vec{b}|^2$

20. If $y = e^{x^2}$, then $\frac{dy}{dx}$ is:

(1) $2xe^{x^2}$ (2) $2x$ (3) e^{x^2} (4) $2x^2e^{x^2}$

PART - II

1. Answer any 7 questions **7 x 2 = 14**
2. Each question carries 2 marks
3. Question number 30 is compulsory

21. Find the middle terms in the expansion of $(x+y)^7$.

22. If the logarithm of 324 to base a is 4, then find a.

23. If two coins are tossed simultaneously, then find the probability of getting (i) one head and one tail (ii) atmost two Tails

24. If $^{15}C_{2r-1} = {}^{15}C_{2r+4}$, find r.

25. Find the distance between the line $4x + 3y + 4 = 0$, and a point $(-2,4)$

26. Determine the value of $x + y$ if $\begin{bmatrix} 2x+y & 4x \\ 5x-7 & 4x \end{bmatrix} = \begin{bmatrix} 7 & 7y-13 \\ y & x+6 \end{bmatrix}$.

27. Find the magnitude of $\vec{a}\times\vec{b}$ if $\vec{a} = 2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{b} = 3\hat{i}+5\hat{j}-2\hat{k}$.

28. Compute (i) : $\lim_{x\to 8}(5x)$ (ii) $\lim_{x\to -2}\left(-\frac{3}{2}x\right)$.

29. Show that the functions $f(x) = \begin{cases} -x+2, & x \le 2 \\ 2x-4, & x > 2 \end{cases}$; $x = 2$ is not differentiable at the indicated value of x.

30. If $y = \sin^2 x$, then find $\frac{d^2y}{dx^2}$

PART - III

1. **Answer any 7 questions** **7 x 3 = 21**
2. **Each question carries 3 marks**
3. **Question number 40 is compulsory**

31. If $f : R - \{-1, 1\} \to R$ is defined by $f(x) = \frac{x}{x^2-1}$, verify whether f is one-to-one or not.

32. A model rocket is launched from the ground. The height h reached by the rocket after t seconds from lift off is given by $h(t) = -5t^2 + 100t$, $0 \le t \le 20$. At what time the rocket is 495 feet above the ground?

33. If the letters of the word IITJEE are permuted in all possible ways and the strings thus formed are arranged in the lexicographic order, find the rank of the word IITJEE.

34. Find the equation of the locus of a point such that the sum of the squares of the distance from the points (3, 5), (1, –1) is equal to 20.

35. If $\cos 2\theta = 0$, determine $\begin{vmatrix} 0 & \cos\theta & \sin\theta \\ \cos\theta & \sin\theta & 0 \\ \sin\theta & 0 & \cos\theta \end{vmatrix}^2$.

36. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 5$ and each one of them being perpendicular to the sum of the other two, find $|\vec{a} + \vec{b} + \vec{c}|$.

37. Suppose that the diameter of an animal's pupils is given by $f(x) = \frac{160x^{-0.4}+90}{4x^{-0.4}+15}$, where x is the intensity of light and f(x) is in mm. Find the diameter of the pupils with (a) minimum light (b) maximum light.

38. Find $f'(3)$ and $f'(5)$ if $f(x) = |x-4|$.

39. Integrate $\frac{1}{(2-3x)^4}$

40. $P(A) = 0.25$; $P(B) = 0.5$ and $P(A\cap B) = 0.15$ then find (i) $P(A\cup B)$, (ii) $P(A\cap\bar{B})$ and (iii) $P(A|\bar{B})$

PART - IV

1. **Answer all the questions** **7 x 5 = 35**
2. **Each question carries 5 marks**

41. a) From the curve $y = x$, draw

(i) $y = -x$ (ii) $y = 2x$ (iii) $y = x + 1$ (iv) $y = \frac{1}{2}x + 1$ (v) $2x + y + 3 = 0$.

(OR)

b) Use the method of undetermined coefficients to find the sum of $1 + 2 + 3 + \cdots + (n-1) + n$, $n \in N$

42. a) In a ΔABC, prove that $\frac{\sin B}{\sin C} = \frac{c - a\cos B}{b - a\cos C}$.

(OR)

b) If $A + B + C = \pi$, prove that $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$.

43. a) Find the sum of all 4-digit numbers that can be formed using digits 0, 2, 5, 7, 8 without repetition?

(OR)

b) Prove that $\sqrt[3]{x^3+6} - \sqrt[3]{x^3+3}$ is approximately equal to $\frac{1}{x^2}$ when x is sufficiently large.

44. a) Express the equation $\sqrt{3}x - y + 4 = 0$ in the following equivalent form:

(i) Slope and Intercept form (ii) Intercept form (iii) Normal form

(OR)

b) Prove that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$.

45. a) Evaluate the following limits: $\lim_{x \to 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x}$

(OR)

b) If ABCD is a quadrilateral and E and F are the midpoints of AC and BD respectively, then prove that $\overrightarrow{AB} + \overrightarrow{AD} + \overrightarrow{CB} + \overrightarrow{CD} = 4\overrightarrow{EF}$.

46. a) Differentiate: $s(t) = \sqrt[4]{\frac{t^3+1}{t^3-1}}$

(OR)

b) Find y'' if $x^4 + y^4 = 16$.

47. a) A tree is growing so that, after t - years its height is increasing at a rate of $\frac{18}{\sqrt{t}}$ cm per year. Assume that when $t = 0$, the height is 5 cm. (i) Find the height of the tree after 4 years. (ii) After how many years will the height be 149 cm?

(OR)

b) A firm manufactures PVC pipes in three plants viz, X, Y and Z. The daily production volumes from the three firms X, Y and Z are respectively 2000 units, 3000 units and 5000 units. It is known from the past experience that 3% of the output from plant X, 4% from plant Y and 2% from plant Z are defective. A pipe is selected at random from a day's total production,

(i) find the probability that the selected pipe is a defective one.

(ii) If the selected pipe is a defective, then what is the probability that it was produced by plant Y?